Causal Effects of a Government Internet Shutdown on Software Production

Evidence from Bangladesh's Nationwide Internet Shutdown in July 2024

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Abstract

This paper estimates the effect of Bangladesh's July 2024 nationwide Internet shutdown on observable software-production activity. I combine public GitHub event data from GH Archive with self-reported user location information and construct a daily panel covering 59,158 pre-active GitHub accounts in Bangladesh and fourteen comparison countries from May 1 to September 4, 2024. The primary outcome is the daily share of eligible users recording productive GitHub activity. A synthetic-control design estimates that during the July 18–23 total-shutdown window, activity in Bangladesh averaged 2.02 percent, compared with a 15.28 percent synthetic counterfactual, an average treatment effect of −13.26 percentage points (−86.8 percent). An international difference-in-differences estimate is nearly identical (−13.09 percentage points). Secondary outcomes show similarly large declines in productive, shipping, and collaborative activity. In-space placebo tests rank Bangladesh as the most extreme unit, although with fourteen placebos the smallest attainable rank p-value is 0.0667. I then use commit-level Git metadata to distinguish code creation from online publication. Among commits whose author matches the Bangladeshi account performing the push, 28.3 percent of commits authored during the shutdown first appear in a public GitHub push on or after July 24, while long push delays are rare in pre-shutdown placebo windows. The evidence therefore indicates a severe collapse in observable online software activity and is consistent with a substantial, but incomplete, margin of local work followed by delayed synchronization. A within-Bangladesh analysis of 20,000 pre-active accounts further shows that recovery after restoration was neither immediate nor uniform across the economic sectors associated with the repositories in which activity occurs: activity in Finance- and Information-related repositories recovered fastest, whereas activity in Accommodation/Food-related repositories remained depressed. Because the shutdown coincided with intense domestic unrest, the estimates

should be interpreted as the effect of the shutdown episode under the assumption that no other Bangladesh-specific shock generated the same abrupt pattern in GitHub activity.

Keywords: Internet shutdowns; software production; GitHub; synthetic control; difference-in-differences; digital economy.

1 Introduction

Internet connectivity is an input into an increasingly large set of production processes. For software developers, the dependence is especially direct: source code can be written locally, but modern development also relies on remote repositories, package registries, code review, issue tracking, continuous integration, documentation, and communication with collaborators. A nationwide loss of connectivity therefore interrupts both the transmission of code and many complementary tasks required to coordinate and ship software.

This paper studies a sharp episode of lost connectivity: Bangladesh's nationwide Internet shutdown in July 2024. Cloudflare reported that mobile networks began disconnecting early on July 18 and that national traffic fell to near zero later that day. Fixed broadband began to return on July 23, initially for selected sectors, while mobile connectivity recovered later, on July 28 (Belson, 2024). Internet Society Pulse records the national disruption as beginning on July 18 and lasting roughly five and a half days (Internet Society, 2024). The event offers an unusual setting in which a digitally intensive form of work can be observed at high frequency before, during, and after a near-complete national loss of Internet connectivity.

The empirical challenge is that GitHub activity is volatile and highly heterogeneous across countries and developers. I therefore construct a multi-country donor pool and use synthetic control as the primary design. The treated unit is Bangladesh, and the donor pool contains fourteen countries with sizable GitHub communities. The synthetic-control weights are chosen using the pre-shutdown path of the daily active-user rate. The resulting counterfactual assigns 47.6 percent weight to Pakistan, 32.5 percent to Ghana, and 19.9 percent to Kenya. During July 18–23, the share of eligible Bangladeshi users with productive GitHub activity averaged only 2.02 percent, compared with 15.28 percent in the synthetic control. The estimated effect is -13.26 percentage points, or -86.8 percent relative to the synthetic counterfactual.

Several complementary designs support the same conclusion. An international difference-in-differences comparison yields an effect of -13.09 percentage points, and excluding national holidays changes the estimate only modestly to -13.38 percentage points. Outcome-specific synthetic controls show declines of 86.2 percent in productive activity, 92.2 percent in shipping activity, and 92.1 percent in collaborative activity. A regression discontinuity in time at the July 17 cutoff produces a smaller but still large discontinuity of roughly -9 percentage points in the daily activity rate.

The paper's second contribution is to distinguish observable online activity from underlying code creation. A Git commit can be created locally without Internet access, while the corresponding push to GitHub requires connectivity. For a large sample of public commits, I recover Git author and committer timestamps and match them to the first observed public PushEvent. In the strict

sample where the commit author is the same GitHub account that performs the push, 829 of 2,932 commits authored during the July 18–23 shutdown, or 28.3 percent, are first observed in a public push on or after July 24. Using the committer timestamp yields 27.2 percent. The share of shutdown-authored commits delayed by more than 24 hours is 29.7 percent, compared with an average of 1.1 percent across six comparable pre-shutdown placebo windows of six days each. These patterns are consistent with local code production continuing for some developers while publication and synchronization were deferred until connectivity returned.

The third part of the analysis turns from the shutdown window to the restoration period. Using a fixed sample of 20,000 Bangladesh-associated accounts selected on pre-shutdown activity, and a text classifier that assigns repositories to NAICS sectors from historical README files, I trace observable push activity by the economic sector associated with the repositories in which it occurs. Restoration of connectivity did not imply an immediate or uniform recovery. Activity in Finance- and Information-related repositories began a sustained run at or above 80 percent of its pre-shutdown level by the third week after the shutdown; Education and Health recovered more gradually; and Accommodation/Food remained well below its baseline throughout the observed window. Activity in pre-existing repositories recovered far less than sector-wide activity, consistent with part of the recovery occurring through new projects. Because this analysis compares each sector with its own pre-shutdown level rather than with an untreated control, it describes heterogeneity in recovery dynamics and does not estimate sector-specific causal effects.

The findings contribute to three literatures. First, they add micro-founded evidence to work on the economic costs of Internet shutdowns. Existing studies document broad effects on output, employment, investment, and economic activity (Tagat, Phokeer, and Kreitem, 2024; Gunsilius and Van Dijcke, 2023). Second, the paper contributes to research on Internet access as a productive input. Hjort and Poulsen (2019), for example, find that the arrival of fast Internet increased employment and firm performance in Africa. Third, the paper joins a growing literature that uses public GitHub activity as a high-frequency measure of software work. Kreitmeir and Raschky (2024) exploit Italy's temporary ChatGPT ban and construct daily GitHub outcomes; Hu et al. (2023) use open-source contributions to study lockdowns; and Wachs et al. (2022) demonstrate the usefulness of GitHub data for measuring the geography of software production.

The interpretation is deliberately narrower than 'total software productivity.' Public GitHub events omit private repositories, other hosting services, meetings, local coding that is never pushed publicly, and non-coding work. User location is self-reported rather than physically verified. Most importantly, the shutdown was implemented amid severe domestic unrest, so the design cannot mechanically separate every contemporaneous Bangladesh-specific shock from the connectivity shock. The paper therefore estimates a large collapse in observable GitHub software-production activity during the shutdown episode, and then uses the commit-timestamp mechanism to examine

whether delayed connectivity is consistent with accounting for part of the observed collapse.

2 Institutional Background

2.1 The July 2024 shutdown

The shutdown followed escalating student protests over public-sector hiring quotas. The political context matters for identification because the decision to restrict connectivity was not randomly assigned. The empirical design therefore relies on the timing and severity of the network disruption, together with cross-country and within-Bangladesh comparisons, rather than assuming that the broader domestic environment was unchanged.

Available network measurements nevertheless provide a precise chronology. Cloudflare reports outages at major mobile operators beginning during the early hours of July 18, followed by a near-complete national loss of Internet traffic around 21:00 local time. Fixed broadband began to recover on July 23, initially as a limited restoration that prioritized sectors including banking, technology, exporters, outsourcing providers, and media. Some residential fixed networks returned later, while major mobile providers did not show substantial recovery until July 28 (Belson, 2024). Internet Society Pulse independently records a national shutdown beginning July 18 and notes the gradual return of connectivity on July 23 (Internet Society, 2024).

The main treatment window in this paper is July 18–23 inclusive. This choice captures the six daily observations most clearly associated with the near-total national disruption while treating July 24 onward as the start of the restoration period in daily data. The distinction is necessarily approximate because restoration began during July 23 and differed across networks. For that reason, specifications based on alternative phase definitions and a regression discontinuity in time are used as robustness checks rather than relying on a single exact daily cutoff.

Table 1: Institutional timeline

Date	Connectivity status	Empirical treatment
July 16–17	Selective social-media/service blocking reported before the nationwide outage	Pre-treatment warning period; July 17 holiday handled in robustness
July 18	Mobile networks disconnect; traffic falls toward zero nationally	Treatment begins
July 18–23	Near-total nationwide Internet disruption	Main total-shutdown window
July 23–27	Partial fixed-broadband restoration	Restoration / post-treatment transition
July 28 onward	Mobile connectivity begins to recover	Post-treatment

Sources: Cloudflare Radar reporting in Belson (2024) and Internet Society Pulse (Internet Society, 2024). Dates are operationalized at daily frequency for the empirical analysis.

2.2 Why GitHub activity is informative about software production

Distributed version control separates code creation from code publication. Developers can create commits locally, but a push to a remote GitHub repository requires network connectivity. Other productive actions—opening or merging pull requests, reviewing code, closing issues, publishing releases, and commenting on collaborative work—are even more directly online. This distinction makes software development useful for studying an Internet shutdown: public GitHub activity measures an observable online margin, while commit metadata can reveal whether some code was created locally and uploaded later.

GH Archive records the public GitHub event stream and makes it available through hourly archives and Google BigQuery (GH Archive, n.d.). The data include events such as PushEvent, PullRequestEvent, PullRequestReviewEvent, IssuesEvent, IssueCommentEvent, and ReleaseEvent. For historical PushEvents, the event payload contains the number of commits and a list of commit objects, subject to a 20-commit timeline-event cap documented by GitHub. I account for this limitation in the mechanism analysis by defining a clean sample of fully observed pushes and by reporting robustness to progressively smaller push sizes (GitHub, n.d.).

3 Related Literature

A first literature estimates the economic consequences of Internet access and Internet disruptions. Hjort and Poulsen (2019) exploit the arrival of submarine Internet cables in Africa and find increases in employment, especially in higher-skill occupations, together with evidence of firm entry, productivity, and exporting. This literature establishes connectivity as a productive infrastructure input rather than merely a consumption amenity. The shutdown literature studies the reverse shock. Tagat, Phokeer, and Kreitem (2024) develop an econometric framework for measuring shutdown costs across countries and emphasize impacts on output, employment, and investment. Gunsilius and Van Dijcke (2023) estimate large reductions in economic activity from an Internet shutdown in India using a discontinuity-based method. These studies motivate the hypothesis that intentional connectivity losses can generate immediate real economic costs.

A second literature uses digital trace data to measure knowledge work. GitHub is especially useful because public events are timestamped and tied to concrete production and collaboration tasks. Wachs et al. (2022) use GitHub to map the geography of open-source software production. Hu et al. (2023) study how pandemic lockdowns affected open-source contributions in China and show that the direction of the effect can depend on context and adaptation. Kreitmeir and Raschky (2024) compile daily GitHub outcomes to study Italy's temporary ban on ChatGPT. Their design is particularly close methodologically: because user-day GitHub actions are sparse and count distri-

butions are highly skewed, they emphasize binary daily indicators, fixed effects, event studies, and robustness to outliers. I follow that logic by making the daily active-user rate the primary outcome and treating raw counts as secondary intensive-margin outcomes.

A third literature concerns comparative-case causal inference. Synthetic control methods construct a weighted combination of untreated units that approximates the treated unit before an intervention (Abadie, Diamond, and Hainmueller, 2010). Abadie (2021) emphasizes transparent donor selection, pre-treatment fit, and placebo or leave-one-out exercises as central diagnostics. Synthetic difference-in-differences combines reweighting of units and time periods with difference-in-differences structure (Arkhangelsky et al., 2021). I use conventional synthetic control as the headline specification because its weights and pre-fit are transparent in this application, while SDID is treated as a robustness exercise. Finally, I use regression discontinuity in time as an additional temporal check, following the cautions in Hausman and Rapson (2018) regarding serial correlation, functional form, and the conceptual differences between time-based and cross-sectional RD designs.

4 Data and Measurement

4.1 GitHub users and country assignment

The analysis begins from GitHub accounts returned by country-location searches for Bangladesh and fourteen potential donor countries: the Philippines, Pakistan, Sri Lanka, Nepal, Indonesia, Vietnam, Malaysia, Thailand, Kenya, Nigeria, Ghana, Morocco, Jordan, and Tunisia. GitHub profile location is self-reported. I therefore interpret the sample as accounts associated with each country rather than verified physical residence at every point in time.

For each account, I use GH Archive to recover public daily GitHub events from May 1 through September 4, 2024. I retain accounts with at least four pre-treatment active days and remove likely bots or automation accounts using login/name patterns and behavioral filters. The final eligible sample contains 59,158 users, including 10,241 accounts associated with Bangladesh. This restriction focuses the design on developers with observable pre-shutdown activity and reduces the influence of dormant accounts. It also means that the estimand is an effect on pre-active public GitHub users, not on every person who has a GitHub account.

4.2 Outcomes

The primary outcome is the daily active-user rate: the share of eligible users in a country who record at least one productive GitHub action on day t. This extensive-margin measure is preferred

because GitHub activity is sparse at the user-day level and raw commit counts contain extreme outliers. Binary activity measures are also closely aligned with the approach used by Kreitmeir and Raschky (2024).

I construct secondary outcomes along the software-production pipeline. Productive activity indicates a day with commits, issue closure, pull-request closure, or a release. Shipping activity indicates a day with a merged pull request or a release. Collaboration activity indicates a day with code review or comment activity. I also construct count outcomes per 1,000 eligible users: collaboration events and shipping output. These measures separate code production from downstream delivery and coordination.

Count outcomes are winsorized in user-level regressions when appropriate, because a small number of accounts generate implausibly large daily or weekly commit counts. The primary country-day synthetic-control analysis does not rely on a raw commit-count headline. National holidays are coded at the country-day level and used in robustness checks.

Table 2: Data and outcome construction

Component	Definition / sample	Scale
Primary outcome	Share of eligible users with any productive GitHub activity	Country-day percentage points
Productive activity	Any commits, issues closed, pull requests closed, or releases	User-day indicator / country-day rate
Shipping activity	Any merged pull request or release	User-day indicator / country-day rate
Collaboration activity	Any PR review, issue comment, PR review comment, or commit comment	User-day indicator / country-day rate
Collaboration events	Total review/comment events	Per 1,000 eligible users
Shipping output	Merged pull requests plus releases	Per 1,000 eligible users
Mechanism sample	Public repo-SHA pairs from fully observed PushEvents, matched to commit metadata	Commit level

Notes: Public GitHub events do not measure private repository activity, work on other platforms, meetings, or local code that is never pushed publicly.

4.3 Commit-level mechanism data

To distinguish coding from publication, I build a separate commit-level dataset for Bangladesh. I extract public PushEvents by eligible Bangladeshi users, retain PushEvents for which the historical payload reveals the complete set of commit SHAs, and keep the earliest observed push for each repository-SHA pair. I then query GitHub commit metadata to recover `authoredDate`, `committedDate`, and GitHub-linked author and committer logins. The analysis target contains

86,065 repository-SHA pairs; 70,576 commit objects are successfully resolved, an 82.0 percent resolution rate.

The strict mechanism sample requires (i) a non-missing author timestamp, (ii) a nonnegative author-to-push delay, and (iii) equality between the GitHub login linked to the commit author and the PushEvent actor. This leaves 59,920 commits. The equality condition avoids attributing a commit written by another person to the Bangladeshi user who happened to push it. Git author and committer timestamps are embedded metadata rather than independently verified work-time measurements, so they are interpreted as evidence consistent with offline production rather than as an infallible clock of when coding occurred.

4.4 Sector-recovery sample and repository classification

To study the period after connectivity returned, I construct a separate within-Bangladesh panel. It starts from the 23,866 Bangladesh-associated accounts with at least one day of selected public GitHub activity between January 18 and July 17, 2024, and draws a fixed sample of 20,000 of them. Eligibility depends only on pre-shutdown activity, so the sample composition cannot respond to the shutdown or to post-shutdown behavior. For these accounts, I recover all public GH Archive events from January 18 through November 23, 2024, aggregated to the user-repository-day level. This sample differs from the 59,158-account synthetic-control sample in its eligibility rule (one rather than four pre-treatment active days), its observation window, and its restriction to Bangladesh, so its activity levels are not directly comparable with the country-day rates in Section 6.

Each repository is assigned to a two-digit sector of the North American Industry Classification System (NAICS) by a publicly released multilingual transformer classifier that predicts a NAICS sector from repository text (`aquiro1994/naics-github-classifier-multilingual`) [CITATION NEEDED]. The classifier input is the repository name and a historical snapshot of the root README file. For repositories observed on or before July 17, 2024, the snapshot is taken as of July 17; for repositories first observed after the shutdown, it is taken as of their first observed activity date. Current repository descriptions and topics are deliberately excluded, because they may have been written after the study period. Of the 168,875 distinct repositories in the panel, a historical README is recovered for 92,979 (55.1 percent); the remainder lack a root README at the target date or could not be retrieved. The main specification assigns a sector only when the classifier's confidence score is at least 0.80, which holds for 32,491 repositories across all sectors; repositories below this threshold do not contribute to any sector. Thresholds of 0.70 and 0.90 are used as robustness checks.

The sector measures the economic sector associated with the repositories in which observable GitHub activity occurs. It is not the developer's employer or industry of employment: the same

developer may contribute to repositories classified in several sectors, and a repository's sector describes the application domain inferred from its text rather than the organization that funds the work.

5 Empirical Strategy

5.1 Synthetic control

The headline design treats Bangladesh as the single treated country and the fourteen other countries as the donor pool. Let Y_{ct} denote the daily active-user rate for country c on date t. Synthetic-control weights are nonnegative, sum to one, and are chosen to minimize the mean squared difference between Bangladesh and the weighted donor combination over the pre-treatment period May 1–July 17. The estimated daily treatment effect is the gap between Bangladesh and the synthetic control during July 18–23.

The method is well suited to the setting because treatment occurs at the country level and the number of aggregate comparison units is modest. Its main advantage here is transparency: the counterfactual is an explicit weighted combination of observed countries, and the credibility of the design can be evaluated through pre-treatment fit, donor weights, in-space placebos, and leave-one-out exercises (Abadie, Diamond, and Hainmueller, 2010; Abadie, 2021).

$$\mathrm{ATT}_{\mathrm{SC}} = \frac{1}{|\mathcal{T}_1|} \sum_{t \in \mathcal{T}_1} \Big[Y_{\mathrm{BD},t} - \sum_{j} w_j Y_{j,t} \Big], \tag{1}$$

where $\mathcal{T}_1$ denotes the shutdown days.

5.2 Difference-in-differences and event study

As a complementary design, I compare the change in Bangladesh's mean outcome from the pre-period to the shutdown with the average corresponding change across the donor countries. Because treatment varies at the country-by-time level, I do not treat user-clustered standard errors from micro regressions as the sole source of inference. The aggregate DiD is used primarily as a magnitude check against the synthetic-control estimate, while placebo and time-series diagnostics provide additional information about whether the Bangladesh break is unusual.

I also estimate user-day fixed-effects specifications and event studies around the shutdown in auxiliary analyses. These models absorb time-invariant user heterogeneity and common date shocks and trace the treatment effect over relative time. The event-study evidence is used diagnostically to assess whether the collapse in activity is concentrated around the treatment window rather than

being the continuation of a smooth pre-existing decline.

$$Y_{it} = \beta\,(\text{Bangladesh}_i \times \text{Shutdown}_t) + \alpha_i + \lambda_t + \text{controls}_{it} + \varepsilon_{it} \tag{2}$$

5.3 Regression discontinuity in time

A regression discontinuity in time at the July 17 cutoff provides a third temporal comparison. I estimate local linear models on daily Bangladesh activity and on a Bangladesh-minus-control residualized series, with heteroskedasticity-and-autocorrelation-consistent standard errors. Following Hausman and Rapson (2018), I vary bandwidths and polynomial order, report donut specifications that omit dates immediately adjacent to the cutoff, inspect autocorrelation, and use placebo cutoffs. Because time itself is the running variable and other contemporaneous events can occur at the same date, the RDiT is interpreted as a robustness check rather than a standalone source of causal identification.

5.4 Inference and robustness

For synthetic control, I implement in-space placebos by assigning the treatment date to each donor country in turn and reconstructing its synthetic control using the remaining donors. The primary rank statistic compares Bangladesh's absolute ATT and post/pre RMSPE ratio with the placebo distribution. With fourteen placebo countries, the smallest possible finite-sample rank value is $1/(14+1) = 0.0667$; consequently, a value of 0.0667 means Bangladesh is the most extreme unit, not that the effect is weak.

I also re-estimate the primary synthetic control after removing each donor country one at a time. Holidays are excluded in a separate DiD specification. Finally, outcome-specific synthetic controls allow donor weights to vary by outcome, whereas the main secondary-outcome table deliberately reuses the active-rate synthetic-control weights. Reusing the primary weights reduces the risk that each new outcome is matched with a different donor combination chosen after observing results.

5.5 Within-Bangladesh sectoral recovery

The designs above estimate the effect of the shutdown during July 18–23. To describe how activity evolved after restoration, I use the fixed sample of 20,000 accounts described in Section 4.4. For user i, sector s, and day t, let $A_{ist} = 1$ if the user records at least one PushEvent on day t in a repository classified to sector s; activity in several repositories of the same sector on the same day counts once. Days are grouped into event periods k: the final eight complete pre-shutdown weeks, $k = -8, \ldots, -1$ (May 23–July 17, 2024); the shutdown, $k = 0$ (July 18–23); and consecutive seven-

day bins after it, starting with $k = +1$ (July 24–30). Let $\bar{A}_{isk}$ be user i's share of active days in period k, and $\bar{A}_{is}^{\text{pre}}$ the corresponding share over the 56 baseline days. I report

$$\beta_{sk} = \frac{1}{N}\sum_{i=1}^{N}\left(\bar{A}_{isk} - \bar{A}_{is}^{\text{pre}}\right), \qquad R_{sk} = 100 \times \frac{\sum_i \bar{A}_{isk}}{\sum_i \bar{A}_{is}^{\text{pre}}}, \tag{3}$$

where $N = 20{,}000$ and users without activity in sector s contribute zeros. The coefficient β_{sk} is the change in the daily probability of sector-s push activity relative to the baseline, in percentage points; the recovery index R_{sk} normalizes the baseline to 100. Standard errors for β_{sk} are the standard deviation of the user-level differences divided by $\sqrt{N}$, and 95 percent intervals use the normal approximation. As a summary of recovery timing, I report the first post-shutdown week that begins two consecutive weeks with $R_{sk} \geq 80$.

Because the activity measure counts pushes to any classified repository, including repositories first observed after the shutdown, it captures recovery through new projects and through switching across repositories. As a complementary specification, I also follow only the repositories pushed to during the baseline weeks, whose activity cannot be replaced by new projects.

This analysis compares each sector with its own pre-shutdown baseline and has no untreated comparison group. It therefore describes the heterogeneity and timing of recovery within Bangladesh; it does not estimate sector-specific causal effects. Post-shutdown deviations from the baseline may also reflect trends, seasonality, or other Bangladesh-specific developments. The confidence intervals capture sampling variation across users but not common shocks that affect all users in a given week. The pre-shutdown coefficients ($k < 0$) indicate how much weekly activity fluctuates around the baseline in the absence of a shutdown and serve as a benchmark for interpreting post-shutdown deviations.

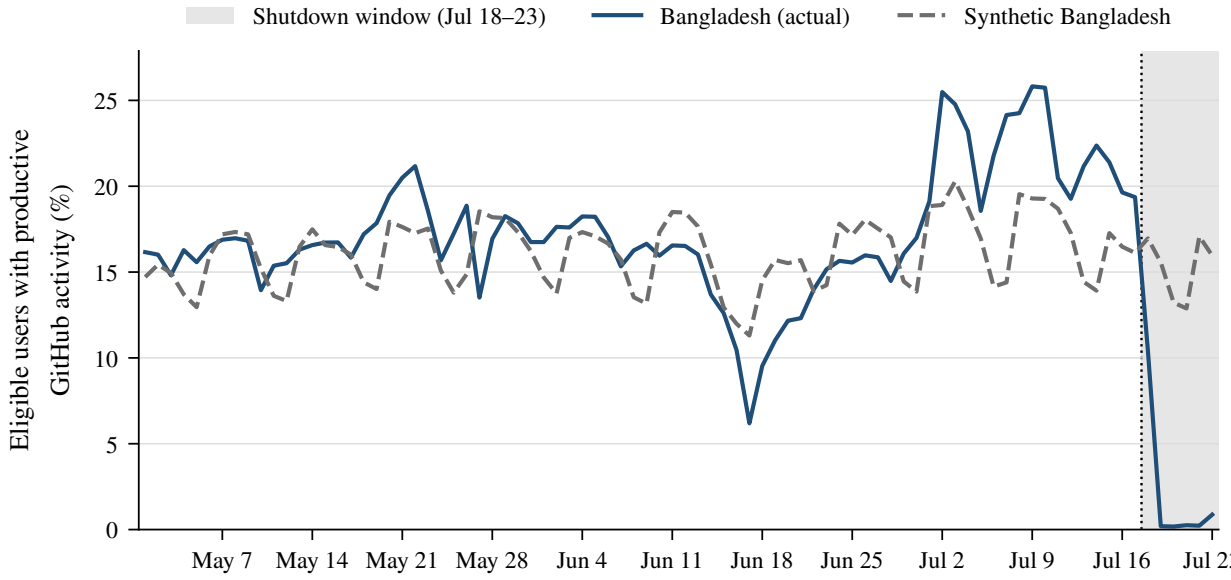
6 Results

6.1 Main effect on daily productive activity

The synthetic-control estimate shows an exceptionally large collapse in observable GitHub activity (Figure 1 and Table 3). Bangladesh's active-user rate averages 2.021 percent during July 18–23, compared with a synthetic counterfactual of 15.277 percent. The ATT is −13.256 percentage points, equivalent to an 86.8 percent reduction relative to the synthetic counterfactual. The pre-treatment RMSPE is 3.260 percentage points and the post/pre RMSPE ratio is 4.192.

The synthetic control places 47.6 percent weight on Pakistan, 32.5 percent on Ghana, and 19.9 percent on Kenya (Table A1). No other donor receives positive weight in the baseline optimization. Bangladesh is more extreme than all fourteen in-space placebos in both absolute ATT and RMSPE-

Figure 1: Actual and synthetic Bangladesh: daily share of eligible users with observable productive GitHub activity

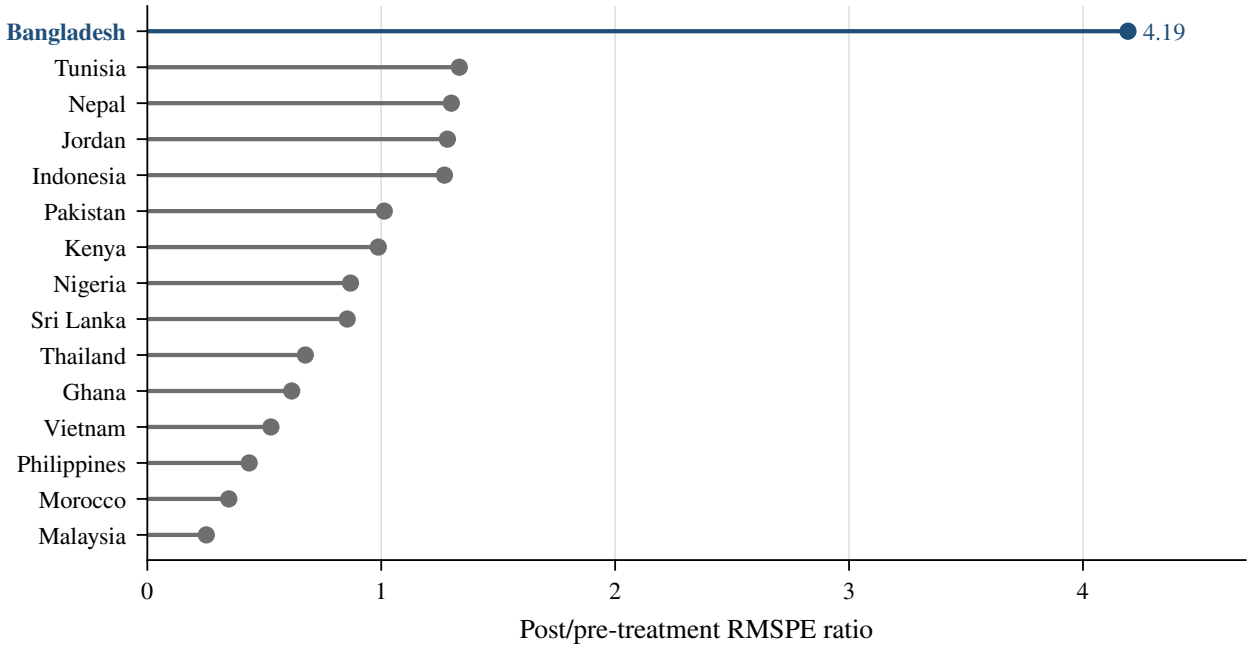


Notes: The solid line is the daily percentage of eligible Bangladesh-associated GitHub accounts with observable productive GitHub activity; the dashed line is the synthetic control, a weighted combination of Pakistan (47.6%), Ghana (32.5%), and Kenya (19.9%) with weights chosen on May 1–July 17, 2024 (Table A1). The shaded band marks the July 18–23 total-shutdown window and the dotted line its onset. The outcome measures observable public GitHub activity, not total software productivity. Pre-treatment fit is imperfect (pre-treatment RMSPE 3.26 percentage points), most visibly in early July, when Bangladesh lies above its synthetic counterpart. The stored daily series ends on July 23, 2024. Source: `results/synthetic_control/actual_vs_synthetic_daily.csv`.

ratio comparisons, yielding rank statistics of 0.0667, the smallest value attainable with fourteen placebo units (Figure 2 and Table A3).

The leave-one-out exercise is highly stable (Table A2). Across specifications that remove one donor at a time, the ATT ranges only from −13.256 to −12.886 percentage points. In particular, excluding Pakistan produces −12.905 percentage points, excluding Ghana produces −13.209 percentage points, and excluding Kenya produces −12.886 percentage points. This matters because both Pakistan and Kenya experienced localized connectivity disruptions near the broader pre-treatment period: Internet Society reports regional restrictions in Pakistan around July 14–17, while Cloudflare documents a major Kenyan traffic disruption on June 25 (Internet Society, 2024; Cloudflare, 2024). The limited movement in the treatment estimate when either donor is removed reduces, though does not eliminate, concern that these donor-specific episodes drive the result.

Figure 2: Synthetic-control in-space placebo inference: post/pre-treatment RMSPE ratios



Notes: Each donor country is assigned the July 18–23 treatment in turn, and its synthetic control is rebuilt from the remaining donors. The figure ranks Bangladesh and the 14 placebo countries by the ratio of post-treatment to pre-treatment RMSPE, a statistic that scales the treatment-period gap by each unit's pre-treatment fit. Bangladesh is the most extreme of the 15 units; it also has the largest absolute ATT. The exact rank p-value is therefore $1/15 = 0.0667$ for both statistics. This is the minimum attainable value with 14 placebo countries: Bangladesh is the most extreme unit in the placebo distribution, but the resolution of exact placebo inference is limited by the size of the donor pool. Source: `results/synthetic_control/sc_inspace_placebos.csv`, verified against `results/final/table_primary_SC_placebos.csv`.

Table 3: Main estimates for the daily active-user rate

Specification	Actual BD (%)	Counter-factual (%)	ATT (pp)	Relative effect	Diagnostic
Synthetic control	2.021	15.277	−13.256	−86.8%	Pre RMSPE 3.260; ratio 4.192
International DiD	2.021	15.109	−13.088	−86.6%	All donor countries
International DiD, no holidays	2.021	15.400	−13.379	−86.9%	National holidays excluded
RDiT, local linear	–	–	−9.284	–	HAC SE 5.034 pp; $p = 0.065$

Notes: The outcome is the daily percentage of eligible users with observable productive GitHub activity; effects are averages over July 18–23, 2024. SC placebo rank values are $1/15 = 0.0667$ for both the absolute-ATT and RMSPE-ratio tests, the minimum attainable with 14 placebo countries (Table A3). The RDiT row is the preferred specification (cutoff July 17, bandwidth 7 days, degree 1, no donut, $N = 15$ days, HAC lags 2) and is treated as a temporal robustness check, not as the main design.

6.2 Difference-in-differences and temporal robustness

The international DiD estimate is −13.088 percentage points, nearly identical to the synthetic-control ATT. Removing national holidays yields −13.379 percentage points. The agreement is notable because the two methods use different weighting logic: synthetic control chooses donor weights to reproduce Bangladesh before the shutdown, whereas the reported international DiD gives equal weight to donor-country changes.

The RDiT produces a smaller point estimate but the same qualitative conclusion. A local linear specification at the July 17 cutoff estimates a discontinuity of −9.28 percentage points in Bangladesh's active-user rate; a controlled Bangladesh-minus-Philippines version yields −9.16 percentage points. HAC p-values are approximately 0.065 in both cases, while a Philippines geographic placebo is near zero (Table A5). Consistent with Hausman and Rapson (2018), I therefore emphasize the direction and timing of the break rather than treating the RDiT p-value as the decisive inferential statistic.

Auxiliary event-study specifications show a decline around the shutdown, but because the archived event-study implementation is not fully documented, I do not use those estimates as primary evidence. Dynamics after restoration are instead documented in Section 6.4, which reports a reproducible within-Bangladesh event study by repository sector.

6.3 Productive, shipping, and collaboration outcomes

The collapse is not limited to a generic activity indicator. Holding the primary synthetic-control weights fixed, productive activity falls by 15.179 percentage points relative to a 17.555 percent counterfactual, or 86.5 percent. Shipping activity falls by 1.333 percentage points relative to a 1.415 percent counterfactual, or 94.2 percent. Collaboration activity falls by 0.780 percentage points relative to a 0.834 percent counterfactual, or 93.6 percent. Count outcomes show similarly large relative declines in collaboration events and shipping output (Table 4).

Allowing the donor weights to re-optimize separately for each outcome produces comparable magnitudes: −86.2 percent for productive activity, −92.2 percent for shipping activity, and −92.1 percent for collaboration activity. The in-space placebo ATT rank is again 0.0667 for all three key activity-rate outcomes. The RMSPE-ratio rank is 0.0667 for productive and shipping activity but 0.20 for collaboration activity. I therefore treat collaboration as consistent secondary evidence rather than giving it the same inferential weight as the primary and shipping outcomes.

Table 4: Secondary outcomes

	Fixed primary weights		Outcome-specific weights		Placebo rank	
Outcome	ATT	Relative	ATT	Relative	ATT	RMSPE ratio
Productive active rate	−15.179 pp	−86.5%	−14.870 pp	−86.2%	0.0667	0.0667
Shipping active rate	−1.333 pp	−94.2%	−0.964 pp	−92.2%	0.0667	0.0667
Collaboration active rate	−0.780 pp	−93.6%	−0.630 pp	−92.1%	0.0667	0.2000
Collaboration events / 1,000	−24.095	−95.1%	−18.589	−93.7%	–	–
Shipping output / 1,000	−23.981	−93.9%	−18.192	−92.1%	–	–

Notes: Fixed-weight specifications reuse the primary active-rate SCM weights (Table A1). Outcome-specific specifications re-estimate donor weights separately for each outcome (Table A4). Rank values are exact in-space placebo ranks for the outcome-specific specifications with 14 placebo countries; 0.0667 is the minimum attainable value. Placebo ranks are not reported for the count outcomes.

6.4 Sectoral recovery after restoration

The synthetic-control estimates describe the collapse during the shutdown window. They do not show how activity evolved once connectivity returned, or whether recovery was common across the kinds of projects on which Bangladeshi developers work. This subsection addresses these questions using the fixed sample of 20,000 Bangladesh-associated accounts and the repository-sector classification of Section 4.4, estimated as described in Section 5.5. The analysis compares each sector with its own pre-shutdown baseline. It documents heterogeneity in recovery dynamics within Bangladesh; it does not estimate sector-specific causal effects relative to an untreated international control, and the synthetic-control estimate in Section 6.1 remains the paper's headline result. Throughout, a sector refers to the economic sector associated with the repositories in which observable GitHub activity occurs, not to developers' employers or industries of employment.

All seven sectors show a sharp fall in observable push activity during July 18–23 (Table 5 and Figure 3). The shutdown-period recovery index ranges from 3.5 in Accommodation/Food to 11.7 in Information and Education and 30.9 in Finance. In every sector, the 95 percent confidence interval for the change relative to the baseline excludes zero (Figure 4).

Restoration of connectivity did not imply an immediate or uniform recovery. In the first week after the shutdown (July 24–30), the recovery index ranges from 33.0 in Real Estate to 92.2 in Finance. Mobile connectivity returned only during this week, later than fixed broadband (Belson, 2024), so part of the week +1 shortfall may reflect incomplete restoration. Recovery was also not monotonic: in Finance, Information, Education, Real Estate, and Accommodation/Food, the index is lower in week +2 (July 31–August 6) than in week +1. The data do not identify the source of this second-week decline.

Finance and Information recovered fastest. Finance returned to an index of 92.2 in week +1, and both sectors begin a run of two consecutive weeks at or above 80 in week +3. Information exceeds

Table 5: Recovery of observable GitHub push activity by repository sector

		Pre	Pre rate	Recovery index (pre-period mean = 100)					First wk
Sector	NAICS	users	(%)	Shutdown	+1	+2	+4	+8	≥80, 2 wks
Finance	52	382	0.078	30.9	92.2	67.6	89.5	91.3	3
Information	51	1,528	0.442	11.7	75.9	72.2	106.0	94.5	3
Education	61	1,234	0.385	11.7	59.1	53.4	75.4	81.9	6
Health	62	514	0.132	6.3	40.7	51.5	77.5	89.4	6
Retail	44–45	1,012	0.295	6.2	52.0	52.3	153.7	79.6	9
Real Estate	53	213	0.063	4.0	33.0	25.1	44.4	85.5	5
Accommodation/Food	72	480	0.144	3.5	44.3	42.8	54.2	62.6	–

Notes: Fixed sample of 20,000 Bangladesh-associated accounts. Repository sectors from the historical-README classifier with confidence ≥ 0.80. The outcome is the share of sampled users with at least one PushEvent to a repository of the sector on a given day. *Pre users*: sampled users with such activity in the baseline period. *Pre rate*: mean daily outcome over the baseline period (May 23–July 17, 2024, the final eight complete pre-shutdown weeks), in percent of all 20,000 users. Shutdown: July 18–23; week +1: July 24–30; subsequent 7-day bins. *First wk ≥80, 2 wks*: first post-shutdown week that begins two consecutive weeks with an index of at least 80 (observed through week +17, November 13–19); it does not imply that the index stays above 80 thereafter. – indicates no such week. Within-Bangladesh comparison with the sector's own pre-period; not a causal effect relative to an untreated control. Sources: `results/sector_recovery/sectorwide/sectorwide_recovery_summary.csv`, verified against `results/sector_recovery/final/table_sector_recovery_main.csv`.

its baseline by week +4 (106.0). In week +8, the two indices are 94.5 and 91.3. The Finance series is nonetheless noisy: its baseline includes a spike in week −1 (index 193.6), and its post-shutdown index falls back below 80 in several later weeks.

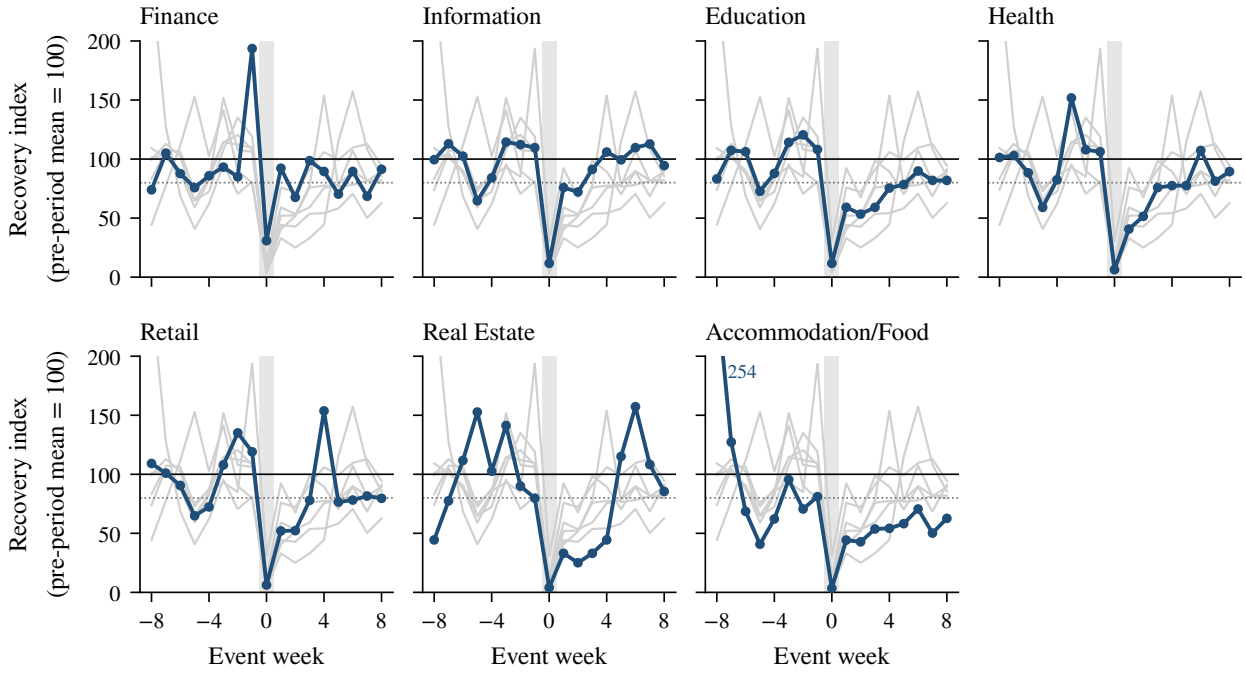
Education and Health recovered more gradually. Their week +1 indices are 59.1 and 40.7, they reach 75.4 and 77.5 by week +4, and both begin a run of two consecutive weeks at or above 80 only in week +6. By week +8, they stand at 81.9 and 89.4.

Accommodation/Food remained persistently depressed. Its index is 44.3 in week +1, 54.2 in week +4, and 62.6 in week +8, and it never records two consecutive weeks at or above 80 through the end of the observation window (week +17, November 13–19). This result should be read alongside the sector's unstable baseline: the index is 254.1 and 127.3 in the first two baseline weeks and between 40.8 and 95.5 in the remaining six. Part of the post-shutdown gap may therefore reflect a decline that began before the shutdown rather than a failure to recover.

Real Estate recovered slowly at first and later approached its baseline. Its index remains between 25.1 and 44.4 through week +4, rises above 100 in weeks +5 and +6 (115.1 and 157.3), and is 85.5 in week +8. This later recovery is not durable within the observed window: from week +12 onward, the index lies between 31.9 and 46.7 (Figure 4). Real Estate is also the smallest of the seven sectors, with 213 sampled users active in the baseline period, so its weekly estimates are imprecise.

Retail shows a pronounced temporary rebound. Its index is about 52 in weeks +1 and +2, rises to

Figure 3: Recovery of observable GitHub push activity by repository sector



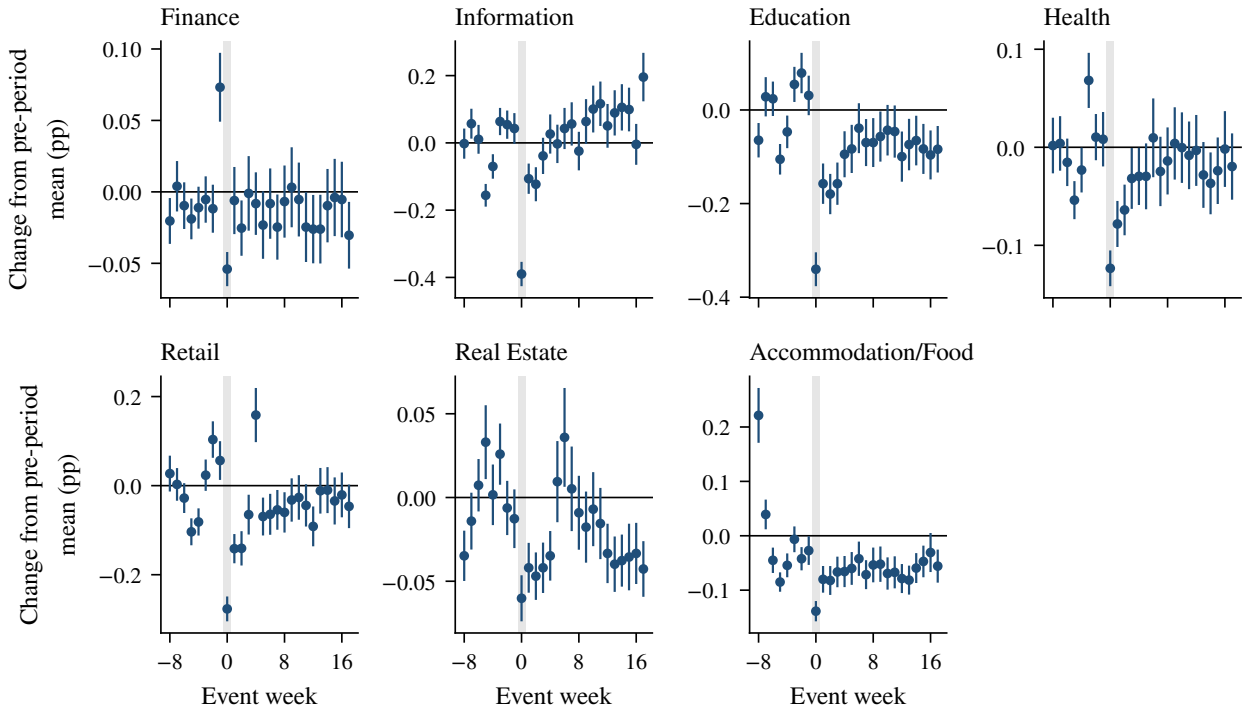
Notes: Each panel highlights one sector's recovery index R_{sk} (Equation (3)); the other six sectors are shown in light grey for comparison. The index is 100 times the sector's mean daily probability of push activity in event week k, divided by its mean over the final eight complete pre-shutdown weeks (May 23–July 17, 2024). The shaded band is the July 18–23 shutdown ($k = 0$); week +1 is July 24–30, followed by consecutive seven-day bins. Solid and dotted horizontal lines mark 100 and 80. The common vertical axis is truncated at 200; the one value above it (Accommodation/Food, week −8) is labeled. Fixed sample of 20,000 Bangladesh-associated accounts; repository sectors from the historical-README classifier with confidence ≥ 0.80. The sector is that of the repositories in which activity occurs, not the developer's employer. Within-Bangladesh comparison with each sector's own baseline; not a causal effect relative to an untreated control. Source: `results/sector_recovery/sectorwide/sectorwide_user_eventstudy.csv`, verified against `results/sector_recovery/final/table_sector_recovery_main.csv`.

153.7 in week +4 (August 14–20), and falls back to 76.5 in week +5. Its first run of two consecutive weeks at or above 80 begins only in week +9. The week +4 value is therefore a transient overshoot, possibly reflecting a burst of catch-up activity, rather than evidence of completed recovery; the data do not identify what generated it. The timing of Retail's first sustained run is also sensitive to the classifier threshold (Section 8.3).

Index values above 100 later in the window should not be read as gains caused by the shutdown. Information, for example, has an index between 111.5 and 126.4 in weeks +10 to +15 and 144.3 in week +17. Without an untreated comparison group, such movements cannot be separated from trend growth, seasonality, or changes in the sampled users' activity.

The main specification counts pushes to any classified repository, including repositories first observed after the shutdown. A complementary specification follows only repositories pushed to during the eight baseline weeks (Table A12). Activity in these pre-existing repositories recovers far

Figure 4: Sectoral event study: change in daily push-activity probability relative to the pre-shutdown baseline



Notes: Points are β_{sk} from Equation (3): the mean change, across the fixed sample of 20,000 Bangladesh-associated accounts, in the daily probability of a PushEvent in a sector-s repository during event week k, relative to each user's mean over the final eight complete pre-shutdown weeks, in percentage points. Vertical lines are 95 percent confidence intervals based on the cross-user standard deviation of these differences; they do not account for common weekly shocks. The shaded band is the July 18–23 shutdown ($k = 0$); week +1 is July 24–30, and the window extends to week +17 (November 13–19, 2024). Vertical scales differ across panels because baseline activity levels differ by sector (Table 5). Repository sectors from the historical-README classifier with confidence ≥ 0.80. Within-Bangladesh comparison; not a causal effect relative to an untreated control. Source: `results/sector_recovery/sectorwide/sectorwide_user_eventstudy.csv`.

less: in week +8, their indices range from 10.1 (Accommodation/Food) to 31.4 (Education), compared with 62.6 to 94.5 in the sector-wide specification. Because repositories selected on recent activity would become less active over time even without a shutdown, this gap does not measure a shutdown effect. It is nevertheless consistent with recovery occurring partly through new repositories and switching across projects rather than solely through a return to pre-existing ones.

Taken together, the sectoral evidence indicates that the collapse in observable activity documented in Section 6.1 was followed by a recovery that was gradual and uneven across the economic sectors associated with developers' repositories. These are before–after comparisons within Bangladesh. They share the identification limitations discussed in Section 8.4 and, in addition, cannot separate post-shutdown recovery from other Bangladesh-specific developments between August and November 2024.

7 Mechanism: Offline Production and Delayed Synchronization

7.1 Main backlog measure

A fall in public GitHub events does not necessarily imply an equal fall in code creation. Git is decentralized: developers may continue editing files and creating commits locally, then push them only after connectivity returns. The commit-level analysis directly tests for this pattern.

In the strict sample, 2,932 commits are authored during July 18–23. Of these, 829 first appear in a public push on or after July 24. The resulting author-time backlog share is 28.27 percent (Table 6). Replacing the author timestamp with the committer timestamp produces a nearly identical 27.16 percent backlog share (790 of 2,909 commits). The agreement between the two metadata clocks is useful because neither timestamp is a perfect external record of when coding occurred. Both timestamps are Git metadata generated by the author's local client or the hosting platform at commit time, not an externally verified record of the exact physical moment the code was written; the mechanism evidence is therefore consistent with offline or local work followed by delayed synchronization, but it does not directly observe the underlying production process.

Long publication delays become dramatically more common during the shutdown. Among shutdown-authored commits, 29.71 percent have an author-to-push delay above 24 hours, 25.38 percent above 48 hours, and 22.75 percent above 72 hours. In six overlapping pre-shutdown placebo windows, the corresponding averages are approximately 1.10 percent, 0.72 percent, and 0.46 percent (Table A6). The real shutdown is therefore far outside the normal pre-period delay distribution.

Table 6: Commit-time mechanism

Timestamp	*N* during shutdown	Pushed Jul 24+	Backlog share	>24h	>48h	>72h	Mean delay (h)
`authoredDate`	2,932	829	28.27%	29.71%	25.38%	22.75%	40.94
`committedDate`	2,909	790	27.16%	28.60%	24.27%	21.73%	37.47

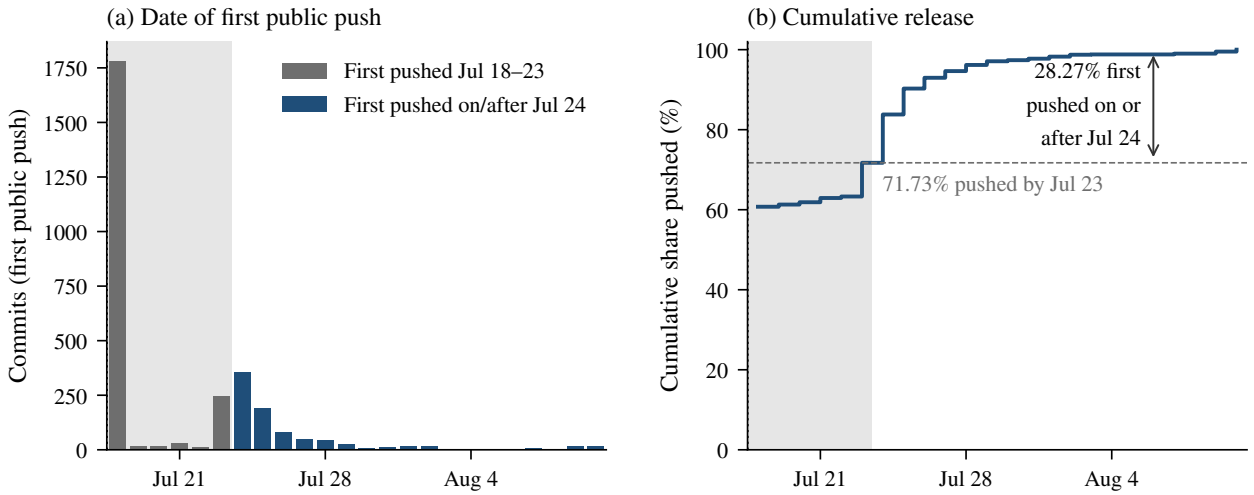
Notes: Strict sample requires author login = PushEvent actor, a non-missing timestamp, and a nonnegative delay. The backlog share is conditional on commits observed in the mechanism sample, not a share of all software production in Bangladesh. `authoredDate` and `committedDate` are Git metadata, not externally verified measurements of when code was written.

7.2 Timing and concentration of delayed pushes

The delayed commits are released quickly once connectivity begins to return (Figure 5). Of the 829 author-time backlog commits, 354 (42.7 percent) appear on July 24 and 544 (65.6 percent) by July

25. By July 27, 81.1 percent have appeared; by July 31, 92.0 percent have appeared. This temporal concentration immediately after restoration is consistent with a synchronization backlog rather than ordinary long-lived branches or sporadic publication.

Figure 5: Release timing of commits with shutdown-period Git author timestamps



Notes: Strict `authoredDate` sample: commits whose GitHub-linked author equals the Bangladeshi PushEvent actor, with a non-missing author timestamp and a nonnegative author-to-push delay. The cohort comprises the 2,932 such commits with an author timestamp between July 18 and July 23, 2024 (calendar days in UTC). Panel (a) shows the date of each commit's first observed public push; panel (b) shows the cumulative share of the cohort that has appeared in a public push. 829 commits (28.27%) first appear on or after July 24, after connectivity began to return. Pushes are observed through August 10, 2024. `authoredDate` and `committedDate` are Git metadata and are not externally verified measurements of the exact physical time code was written; the pattern is consistent with offline or local work followed by delayed synchronization. The share refers only to this observable strict commit sample, not to all software production in Bangladesh. Source: `results/mechanism/commit_sha_author_push_master.parquet`, verified against `results/final/table_mechanism_main.csv` and `results/mechanism/backlog_release_dates.csv`.

The backlog is not generated by only a handful of unusually active users (Table A10). The 2,932 shutdown-authored commits come from 835 users and 977 repositories. Two hundred twenty-two users (26.6 percent) and 246 repositories (25.2 percent) contribute at least one backlog commit. The largest user accounts for 2.77 percent of backlog commits, the top five users for 12.06 percent, and the top ten for 21.47 percent. Dropping the twenty users with the most backlog still leaves a 20.37 percent backlog share.

Backlog is, however, more common in larger pushes (Table A9). Restricting to pushes with at most 10 commits lowers the backlog share to 21.0 percent; at most 5 commits, 14.4 percent; at most 2 commits, 7.8 percent; and single-commit pushes, 3.3 percent. This pattern is economically meaningful rather than merely inconvenient: if developers accumulate local commits while offline and synchronize later, restoration should produce larger-than-normal push batches. At the same time, it means that the 28.3 percent headline mechanism estimate should not be interpreted as

applying uniformly to every kind of push.

8 Robustness, Identification Threats, and Limitations

8.1 Synthetic-control diagnostics

The primary synthetic control improves pre-treatment fit relative to using the Philippines alone, but the fit is not perfect. The pre-treatment RMSPE is 3.26 percentage points, and Bangladesh exhibits temporary deviations from the synthetic series in early July (Figure 1). The paper therefore reports pre-fit explicitly rather than treating the synthetic control as a perfect replica. The unusually large post-treatment gap and the stability of the estimate under leave-one-out exercises are the main reasons the design remains informative.

The donor pool also contains countries with their own short-lived connectivity disturbances. Pakistan experienced regional restrictions around July 14–17, immediately before Bangladesh's national shutdown, and Kenya experienced a major traffic disruption on June 25. These episodes are a reason to avoid presenting donor selection as mechanically pristine. Still, removing Pakistan or Kenya individually changes the ATT by less than 0.4 percentage points. A more conservative future extension could pre-register donor exclusion rules based on third-party connectivity data and expand the donor pool to increase placebo-inference resolution.

8.2 Metadata and measurement

Commit metadata resolution is similar across periods: 82.34 percent of the pre-period target commits are found, compared with 84.24 percent during the shutdown and 81.81 percent after (Table A7). The shutdown mechanism is therefore not generated by a sharp deterioration in commit-object availability during the treatment window.

A related limitation concerns push-listing completeness: although 99.5–99.8 percent of pushes in every period expose a complete commit list, the share of nominal commits captured within complete pushes falls from 90.27 percent before the shutdown to 73.11 percent during the shutdown and 71.45 percent afterward (Table A8). This occurs because a small number of large pushes account for a disproportionate share of commits and are more likely to have incomplete commit listings. Since the strict mechanism sample restricts to complete pushes, this does not mechanically invalidate the 28.27 percent backlog estimate, but it does imply uneven commit-level coverage across periods. The lower coverage during and after the shutdown is therefore a measurement limitation that should be considered when interpreting the mechanism results.

Location is another measurement limitation. GitHub location fields are user-entered and may

be missing, stale, or imprecise. The estimates should be read as applying to accounts associated with Bangladesh and the donor countries, not as a verified census of developers physically located in each country on every date. Public-event data additionally exclude private repositories. If private and public work respond differently to a shutdown, the paper estimates the effect on the observable public margin rather than on all software production.

8.3 Sector classification and the sectoral recovery analysis

Table A11 repeats the sectoral recovery analysis keeping only repositories whose classifier confidence is at least 0.70 or 0.90 instead of 0.80. The qualitative ordering of the main sectors is stable. At all three thresholds, Finance and Information begin their first run of two consecutive weeks at or above 80 in week +3; Education and Health do so between weeks +5 and +8; and Accommodation/Food never does, with week +8 indices between 60.3 and 64.9. Real Estate's late recovery is broadly but not fully robust: its week +8 index is 90.0 and 85.5 at the 0.70 and 0.80 thresholds but 75.5 at 0.90. Retail's week +4 overshoot appears at every threshold (indices between 153.7 and 159.1), but the start of its first two-week run at or above 80 varies from week +4 (0.70) to week +9 (0.80) and week +7 (0.90). I therefore do not emphasize a precise recovery date for Retail.

Three sectors with substantial activity are excluded from the main results. Professional/ Scientific and Administrative Support have 577 and 196 sampled users active in the baseline period at the 0.80 threshold, but only 125 and 48 at 0.90, so their indices depend heavily on the threshold. In Arts, Entertainment, and Recreation, a single repository accounts for 82.5 percent of pre-shutdown events among the sector's classified repositories, so a sector index would largely track one project.

The classification has further limitations. First, the sector describes the application domain inferred from a repository's text, not the employer or industry of the developer: a developer employed in banking may work on a general-purpose library, and a student exercise may be classified as Retail because it implements an online shop. Second, classifier errors are possible, and this paper does not validate the classifier's accuracy on Bangladeshi repositories. Third, coverage is incomplete. Only 55.1 percent of repositories have a recoverable historical README, and activity in unclassified repositories does not enter any sector. If newly created repositories are less likely than established ones to have a README at their first observed activity date, post-shutdown recovery through new projects would be understated. Fourth, a user may contribute to several sectors, so sector indices are not shares of a fixed workforce. Finally, the sectoral confidence intervals do not adjust for the many sector-week comparisons reported, and they reflect sampling variation across users rather than common shocks.

8.4 Concurrent unrest and causal interpretation

The most important identification threat is that the Internet shutdown was itself a response to political unrest. Protests, curfews, transportation disruptions, workplace closures, fear, or other domestic conditions could directly reduce software work even if connectivity were available. Date fixed effects and foreign controls cannot absorb shocks unique to Bangladesh that coincide with the shutdown. For this reason, the headline estimate should be interpreted as the effect of the shutdown episode under the identifying assumption that the abrupt Bangladesh-specific deviation is primarily attributable to lost connectivity.

Several findings make the connectivity interpretation more plausible without fully proving it. First, the break is tightly aligned with the network outage. Second, activity begins to recover as connectivity returns. Third, downstream online tasks such as shipping and collaboration fall at least as strongly as the general activity rate. Fourth, commit metadata reveal a large stock of code authored during the outage but pushed immediately after restoration. A pure 'developers stopped working because of unrest' story does not naturally predict that last pattern. Nevertheless, the mechanism evidence is complementary, not a substitute for randomized assignment.

9 Discussion

The results illustrate why high-frequency digital trace data can add information beyond aggregate estimates of shutdown costs. A national GDP series cannot show whether digital workers cease production, shift to offline tasks, or accumulate work that is released later. GitHub timestamps reveal all three margins imperfectly but at daily and commit-level frequency.

The main estimate—an 86.8 percent decline in the daily active-user rate relative to the synthetic counterfactual—is best interpreted as a collapse in observable online software-production activity. The mechanism results indicate that some of this activity is displaced across time rather than destroyed. At least within the strict public-commit sample, roughly one-quarter to one-third of commits authored during the outage appear only after the main shutdown window. This implies that treating observed pushes as identical to contemporaneous production would overstate the immediate loss in underlying coding effort.

At the same time, delayed synchronization cannot account for the entire collapse. Most eligible users record no productive public activity during the shutdown, and the recovery backlog is much smaller than the counterfactual volume implied by the active-rate and shipping estimates. Connectivity therefore appears to matter not only for publishing already-written code but also for complementary tasks: accessing remote dependencies, coordinating with collaborators, reviewing code, resolving issues, merging pull requests, and releasing software. The especially large declines

in shipping and collaboration are consistent with these channels.

The sectoral results add a dynamic dimension. Reconnection did not restore observable activity at once: weeks after the shutdown, activity in Education-, Health-, and especially Accommodation/Food-related repositories remained below its pre-shutdown level, while Finance- and Information-related activity recovered within about three weeks. This pattern is consistent with the costs of a shutdown extending beyond the days without connectivity and falling unevenly across types of projects. The weaker recovery of pre-existing repositories, relative to sector-wide activity, further suggests that part of the recovery took the form of new projects or switching across repositories, which an analysis restricted to pre-existing projects would miss. Because the sectoral analysis has no untreated comparison group and classifies repositories rather than workers, these interpretations remain suggestive.

These results are relevant for firms and policymakers in digitally intensive economies. Resilience planning that focuses only on local editing tools may be insufficient because modern software production depends on a broader stack of remote services. Local mirrors of repositories and package registries, redundant international connectivity, offline-capable documentation, and asynchronous development workflows may mitigate some losses, but they cannot fully substitute for network access when teams, deployment infrastructure, and customers remain online.

10 Conclusion

Bangladesh's July 2024 Internet shutdown provides a high-frequency case study of what happens to digitally mediated production when national connectivity is abruptly removed. Using public GitHub events for 59,158 pre-active accounts across fifteen countries, the paper finds that Bangladesh's daily productive-activity rate fell by 13.26 percentage points relative to a synthetic counterfactual during July 18–23, an 86.8 percent decline. International difference-in-differences, secondary outcomes, placebos, and temporal discontinuity analyses point in the same direction.

The commit-level evidence adds an important qualification. A substantial share of commits authored during the outage appeared only after connectivity began to return. These patterns suggest that observable online activity fell more sharply than locally recorded code creation within the strict public-commit sample. This distinction is central to measuring the effects of connectivity shocks on observable software-development activity: an outage can reduce contemporaneous online activity while shifting part of the remaining work into an offline backlog that is synchronized later.

A within-Bangladesh analysis of the restoration period adds a further qualification: reconnection did not immediately restore observable activity. Recovery was faster in Finance- and Information-related repositories than in Education- and Health-related repositories, and activity in Accommodation/Food-related repositories remained depressed throughout the observed window.

These sectoral patterns are descriptive comparisons with each sector's own pre-shutdown level rather than causal estimates against an untreated control.

The paper's main limitation is the coincidence of the shutdown with severe domestic unrest. Future work could combine network telemetry at subnational or provider level with developer location at finer geographic resolution, or exploit variation in restoration timing across networks and regions. Such designs could more sharply separate the direct effect of connectivity from the broader social environment. Even with this limitation, the evidence is consistent with observable software-development activity being highly exposed to network availability and with public development platforms providing unusually granular measures of that exposure.

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A Additional Empirical Diagnostics

This appendix reports the detailed tables behind the synthetic-control, RDiT, mechanism, and sectoral recovery results discussed in the main text. All tables are generated from the stored result files by `analysis/tables/make_all_tables.py`.

A.1 Synthetic control

Table A1: Primary synthetic-control donor weights

Country	Weight
Pakistan	47.579%
Ghana	32.488%
Kenya	19.933%

Notes: Weights are estimated on the pre-treatment (May 1–July 17, 2024) daily active-user rate. The remaining eleven donor countries receive zero weight.

Table A2: Primary leave-one-out range

Donor removed	ATT (pp)	Relative effect	Pre RMSPE	RMSPE ratio
Ghana	−13.209	−86.73%	3.297	4.131
Pakistan	−12.905	−86.46%	3.376	3.961
Kenya	−12.886	−86.44%	3.293	4.042
Any zero-weight donor (11)	−13.256	−86.77%	3.260	4.192

Notes: Removing donors that receive zero baseline weight mechanically leaves the baseline estimate unchanged.

Table A3: Synthetic-control in-space placebo inference

Rank	Unit	ATT (pp)	Pre RMSPE	Post RMSPE	Ratio
1	**Bangladesh**	−13.256	3.260	13.666	4.192
2	Tunisia	−2.563	2.011	2.682	1.334
3	Nepal	0.965	0.936	1.216	1.300
4	Jordan	−0.194	2.457	3.151	1.283
5	Indonesia	−1.393	1.151	1.462	1.270
6	Pakistan	0.941	1.585	1.606	1.013
7	Kenya	2.439	2.873	2.837	0.987
8	Nigeria	0.527	0.948	0.823	0.869
9	Sri Lanka	−0.748	1.382	1.181	0.855
10	Thailand	−0.215	0.965	0.652	0.675
11	Ghana	0.268	1.144	0.706	0.617
12	Vietnam	−0.097	0.867	0.458	0.528
13	Philippines	0.091	0.955	0.416	0.435
14	Morocco	0.050	1.794	0.625	0.348
15	Malaysia	0.131	1.177	0.297	0.252

Notes: Each donor country is assigned the July 18–23 treatment in turn and its synthetic control is rebuilt from the remaining donors. Units are ordered by the post/pre RMSPE ratio. Bangladesh has both the largest ratio and the largest absolute ATT among the 15 units, so the exact rank p-value is $1/15 = 0.0667$ for both statistics. This is the minimum attainable value with 14 placebo countries; inference resolution is limited by the size of the donor pool.

Table A4: Outcome-specific synthetic-control donor weights

Outcome	Country	Weight
Productive active rate	Pakistan	36.1%
	Philippines	25.2%
	Jordan	21.3%
	Kenya	16.5%
	Thailand	0.8%
Shipping active rate	Pakistan	81.8%
	Ghana	8.9%
	Morocco	7.2%
	Jordan	2.2%
Collaboration active rate	Pakistan	92.3%
	Morocco	7.7%
Collaboration events / 1,000	Morocco	84.7%
	Nigeria	10.1%
	Ghana	5.2%
Shipping output / 1,000	Pakistan	72.5%
	Morocco	8.4%
	Ghana	8.2%
	Nigeria	7.8%
	Philippines	2.3%
	Jordan	0.8%

Notes: Donors with zero weight are omitted. These weights are reported for transparency; the main secondary-outcome comparison holds the primary weights in Table A1 fixed.

A.2 Regression discontinuity in time

Table A5: Regression discontinuity in time: preferred specification

Specification	Outcome	Effect (pp)	HAC SE	95% CI	p	Days
Main local linear	Bangladesh active rate	−9.284	5.034	[−19.151, 0.584]	0.065	15
Controlled local linear	Bangladesh − Philippines active rate	−9.160	4.980	[−18.921, 0.601]	0.066	15
Geographic placebo	Philippines active rate	−0.124	1.492	[−3.048, 2.801]	0.934	15

Notes: Local linear RDiT at the July 17, 2024 cutoff, bandwidth 7 days, polynomial degree 1, no donut, HAC lags 2. Effects are reported in percentage points. RDiT is a temporal robustness check, not the primary causal design. Source: `results/rdit/main_rdit_summary.csv`.

A.3 Mechanism

Table A6: Mechanism placebo comparison

Metric	Real shutdown	Mean placebo	Maximum placebo
Released in next 6 days	25.38%	0.56%	0.77%
Delay >24h	29.71%	1.10%	1.31%
Delay >48h	25.38%	0.72%	0.90%
Delay >72h	22.75%	0.46%	0.56%
Delay >120h	13.88%	0.32%	0.46%

Notes: Placebo windows are six six-day authoring windows between July 1 and July 11, with six-day subsequent release windows that end before treatment. Windows overlap, so the comparison is descriptive rather than an exact independent randomization test.

Table A7: Metadata coverage by push period

Period	Target N	Commit found	Author timestamp	Author login	Author = pusher
Pre	17,000	82.34%	82.34%	77.95%	71.49%
Shutdown	3,197	84.24%	84.24%	79.17%	69.69%
Post	65,868	81.81%	81.81%	77.81%	71.05%

Notes: Similar resolution rates across periods reduce concern that the mechanism result is generated by treatment-specific metadata missingness.

Table A8: Push-listing completeness by period

Period	Pushes	Mean push size	Complete pushes	Nominal commits	Commits in complete pushes
Pre	121,960	1.29	99.78%	157,834	90.27%
Shutdown	2,738	1.70	99.53%	4,667	73.11%
Post	58,110	1.69	99.66%	98,022	71.45%

Notes: A push is complete when its historical payload lists every commit. Large pushes account for a disproportionate share of commits and are more likely to be incomplete. Source: `results/mechanism/robust_push_size_period.csv`.

Table A9: Mechanism estimates by push size

Sample	N	Backlog N	Backlog share	>24h	>48h	>72h
All complete pushes	2,932	829	28.27%	29.71%	25.38%	22.75%
Push size ≤ 10	2,503	526	21.01%	21.81%	18.98%	17.14%
Push size ≤ 5	2,217	320	14.43%	15.07%	13.62%	12.45%
Push size ≤ 2	1,947	151	7.76%	7.96%	7.29%	6.93%
Single-commit pushes	1,707	57	3.34%	3.28%	2.93%	2.75%

Notes: Strict `authoredDate` sample of commits authored July 18–23. The backlog share is not invariant to push size.

Table A10: Concentration of the backlog across users and repositories

Statistic	Value
Shutdown-authored commits	2,932
Backlog commits	829 (28.27%)
Users / with backlog	835 / 222 (26.59%)
Repositories / with backlog	977 / 246 (25.18%)
Largest user's share of backlog	2.77%
Top 5 users' share of backlog	12.06%
Top 10 users' share of backlog	21.47%
Backlog share after dropping top-k backlog users	
$k = 1$ ($N = 2,909$)	27.71%
$k = 3$ ($N = 2,868$)	26.67%
$k = 5$ ($N = 2,831$)	25.75%
$k = 10$ ($N = 2,739$)	23.77%
$k = 20$ ($N = 2,577$)	20.37%

Notes: Strict `authoredDate` sample. Users are ranked by their number of backlog commits.

A.4 Sectoral recovery

Table A11: Sectoral recovery indices by classifier-confidence threshold

Sector	Threshold	Pre users	Week +1	Week +4	Week +8	First wk ≥80, 2 wks
Finance	0.70	437	91.6	92.4	89.2	3
	0.80	382	92.2	89.5	91.3	3
	0.90	275	101.7	82.5	101.7	3
Information	0.70	2,123	73.0	116.6	101.3	3
	0.80	1,528	75.9	106.0	94.5	3
	0.90	968	79.5	97.6	91.5	3
Education	0.70	1,345	61.6	75.4	79.4	5
	0.80	1,234	59.1	75.4	81.9	6
	0.90	1,029	57.8	73.3	77.0	6
Health	0.70	542	40.8	76.4	88.3	6
	0.80	514	40.7	77.5	89.4	6
	0.90	455	41.1	75.5	87.6	8
Retail	0.70	1,090	52.2	159.1	82.5	4
	0.80	1,012	52.0	153.7	79.6	9
	0.90	829	50.1	157.4	80.9	7
Real Estate	0.70	228	33.1	45.5	90.0	5
	0.80	213	33.0	44.4	85.5	5
	0.90	153	44.9	44.9	75.5	5
Accommodation/Food	0.70	514	46.0	59.3	64.9	–
	0.80	480	44.3	54.2	62.6	–
	0.90	394	39.2	45.1	60.3	–
Excluded from the main figures and table						
Professional/Scientific	0.70	1,241	57.9	66.9	91.2	7
	0.80	577	64.9	64.9	91.0	7
	0.90	125	94.4	65.6	78.7	1
Admin. Support	0.70	268	30.8	83.0	77.0	9
	0.80	196	32.2	72.9	84.7	10
	0.90	48	5.2	61.9	56.8	–

Notes: Each row repeats the main specification (Table 5) keeping only repositories whose classifier confidence is at least the stated threshold; 0.80 is the main specification. Professional/Scientific and Administrative Support are excluded from the main results because their support falls sharply at the 0.90 threshold. Source: `results/sector_recovery/final/sector_threshold_robustness_full.csv`.

Table A12: Recovery of pre-existing repositories versus sector-wide user activity

		Fixed pre-period repositories			Sector-wide (main)		
Sector	Repos	+1	+4	+8	+1	+4	+8
Finance	417	60.0	28.3	23.0	92.2	89.5	91.3
Information	2,040	46.8	37.4	27.3	75.9	106.0	94.5
Education	1,523	40.3	38.2	31.4	59.1	75.4	81.9
Health	651	25.1	28.6	23.1	40.7	77.5	89.4
Retail	1,322	28.7	30.9	19.8	52.0	153.7	79.6
Real Estate	254	16.1	25.7	30.0	33.0	44.4	85.5
Accommodation/Food	581	29.4	16.1	10.1	44.3	54.2	62.6

Notes: Recovery indices (pre-period mean = 100). *Fixed pre-period repositories*: classified repositories (confidence ≥ 0.80) with at least one PushEvent by a sampled user during the final eight pre-shutdown weeks; the outcome is each repository's daily PushEvent active-day rate, and new repositories never enter. *Sector-wide*: main specification of Table 5, which includes activity on repositories first observed after the shutdown. Because a cohort selected on recent activity would decline over time even without a shutdown, the fixed-repository indices are not a counterfactual for the sector-wide indices. Sources: `results/sector_recovery/pre8/sector_recovery_summary_pre8.csv` and `results/sector_recovery/sectorwide/sectorwide_recovery_summary.csv`.